

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE/IT)

November 2012

IT303 Advanced Database Management System

Date: 29.11.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) List out the guidelines to create a view. Can a view based on another view ? Justify you answer. [04]
- (b) Explain the Levels of Data and Process Distribution in DDAS environment. [03]
- Q - 2 (a) Explain, the meaning and purpose of compensating transactions. Give two examples of their use. [06]
- (b) Transaction server architectures are popular for client-server relational databases, where transactions are short. On other hand, data server architectures are popular for client server object oriented databases systems, where transactions are expected to be relatively long. Give two reasons why data servers may be popular for object oriented databases but not for relational databases. [05]
- (c) How Procedures and Functions Differ? Explain with an example. [03]

OR

- Q - 2 (a) How does the concept of an object in the object-oriented model differ from concept of an entity in the entity-relationship model? Illustrate with proper example. [06]
- (b) What is tablespace? Can objects of the same schema reside in different tablespaces? Justify your answer. [05]
- (c) What is a cursor ? Explain parameterized cursor. [03]
- Q - 3 (a) A B+ tree of order 4 (no. of pointers $N=4$) is built from scratch by 10 successive insertions. [08]
- (1) What is the maximum number of node splitting operations that may take place?
- (2) Insert 12,14 and 16 in original tree and modify
- (3) Final tree after deleting 3,9 from original tree.
- (b) Distinguish between public and private keys in an asymmetric key cryptosystem. Compare and contrast the keys in symmetric key and asymmetric key cryptosystems. [06]

OR

Q - 3 (a) Answer the following:

[08]

- (1) Compare and contrast coarse granularity and fine granularity.
- (2) Given a list of identifier in the following:

Identifiers	Binary representation
A0	100 000
A1	100 001
B0	101 000
B1	101 001
C0	110 000
C1	110 001
C2	110 010
C3	110 011
C5	110 101

Now put these identifiers into a table of four pages. Each page can hold at most two identifiers, and each page is indexed by two-bit sequence 00, 01, 10, 11.
Construct binary tree (Trie) using dynamic hashing.

(b) What Are the Key Elements of Parallel Processing? Explain any one of them in detail. [06]

SECTION – II

Q - 4 (a) Attempt the following:

[04]

- (1) Differentiate static hashing and dynamic hashing.
- (2) How PL/SQL Exceptions Are Raised?

(b) Explain about Clustered and non clustered index? How to choose between a Clustered Index and a Non-Clustered Index? [03]

Q - 5 (a) What is distributed database? Explain two-phase commit (2 PC) protocol in distributed database. [06]

(b) Refer the following given tables and SQL query:

[04]

DEPT	EMP	
DEPT#	EMP#	DEPT#
D2	E1	UNK

SQL query: select emp# from dept, emp where not (dept.dept# = emp.dept# and emp.dept# = 'D1')

Evaluate the above SQL query (i) without any optimization (ii) with optimization (predicate transitive closure).

(c) Draw detail structure of TP monitor and explain in detail roles of various manager. [04]

OR

Q - 5 (a) How database snapshots works ? Explain with proper diagram.

[05]

- (b) What form of parallelism (interquery, interoperation, or intraoperation) is likely to be the most important for each of the following tasks. [05]
- Increasing the throughput of a system with many small queries
 - Increasing the throughput of a system with a few large queries, when the number of disks and processors is large.
- (c) Define the following terms: [04]
- Data file
 - Instance
 - PMON
 - Redo log buffer

- Q - 6 (a) Answer the following (Any Three) [12]
- What is NULL value? How to handle NULL values?
 - Explain supertypes and subtypes in type hierarchy.
 - If an object is created without any reference to it, how can that object be deleted?
 - I/O parallelism refers to reducing the time required to retrieve relation from disk by partitioning the relations on multiple disk. In contrast to it, list out which are the partitioning techniques used and explain it in detail.
- (b) Explain the concept of Row Chaining. [02]
